AI-Powered Agricultural Decision Support System with CROP CLASSIFICATION AND YIELD PREDICTION

—

Using CNN, ML (LGBM), RAG, LLM models with remote sensing satellite images, table, text dataset

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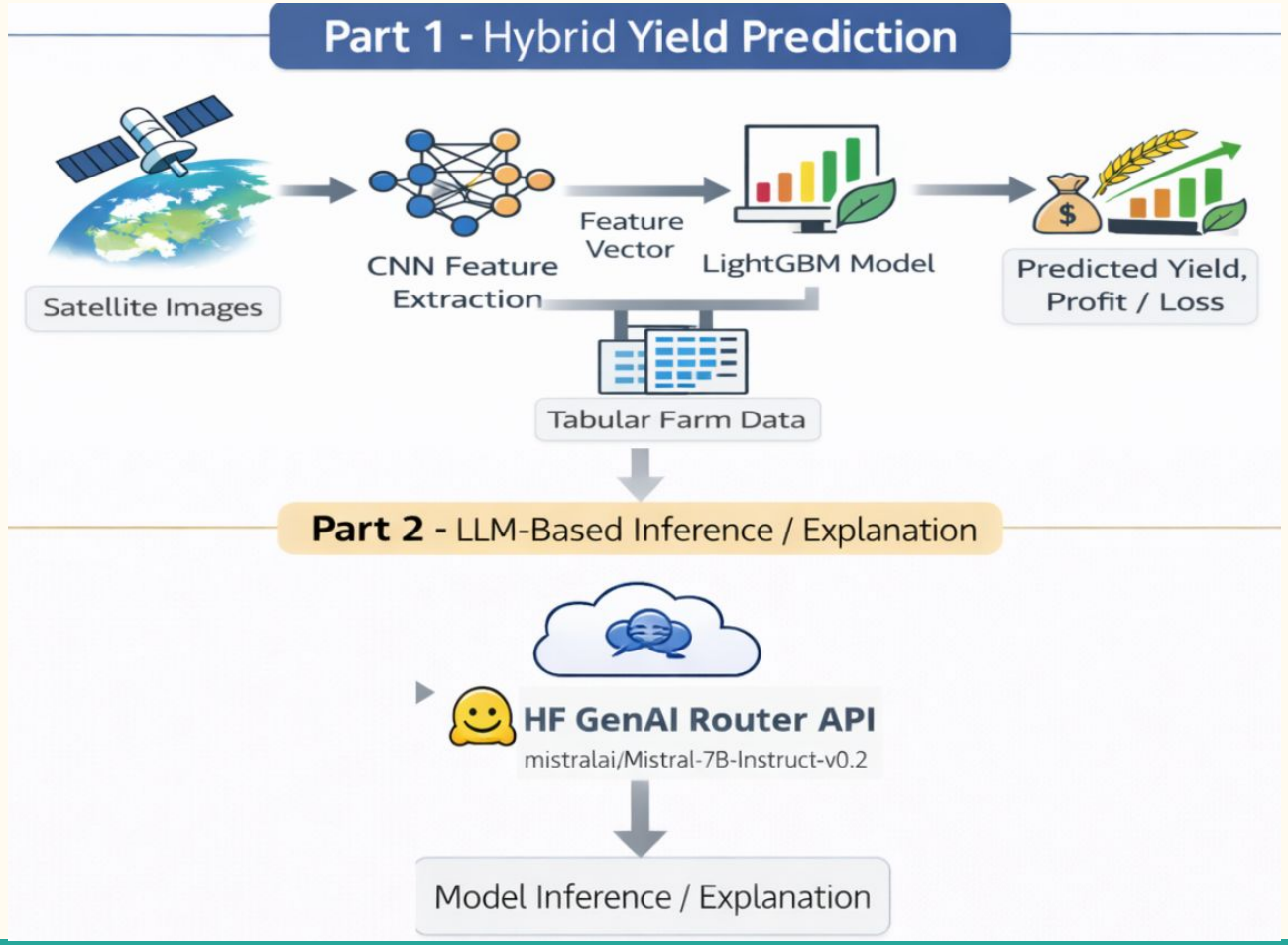
🎯 Aim

To develop an end-to-end AI-driven agricultural decision support platform integrating satellite imagery, tabular farm data, **machine learning**, **deep learning** and **generative AI**.

🎯 Objectives

- Extract vegetation features from satellite images using **CNN**
- Perform hybrid yield prediction using **LightGBM**
- Estimate profit/loss based on predicted yield
- Generate explainable agronomic advisory using **LLM**
- Build a **RAG**-based agricultural QA system using **FAISS** and **Qwen** 1.8B

Project Workflow



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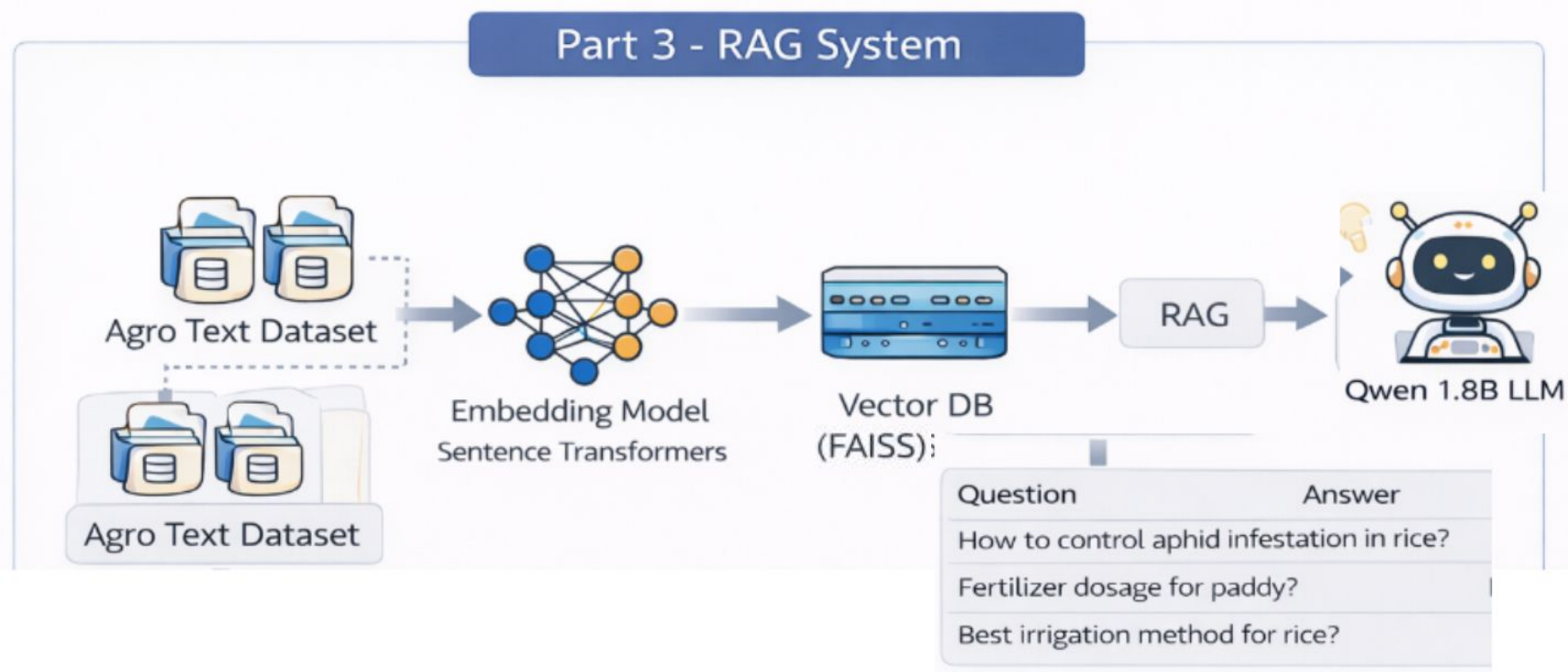
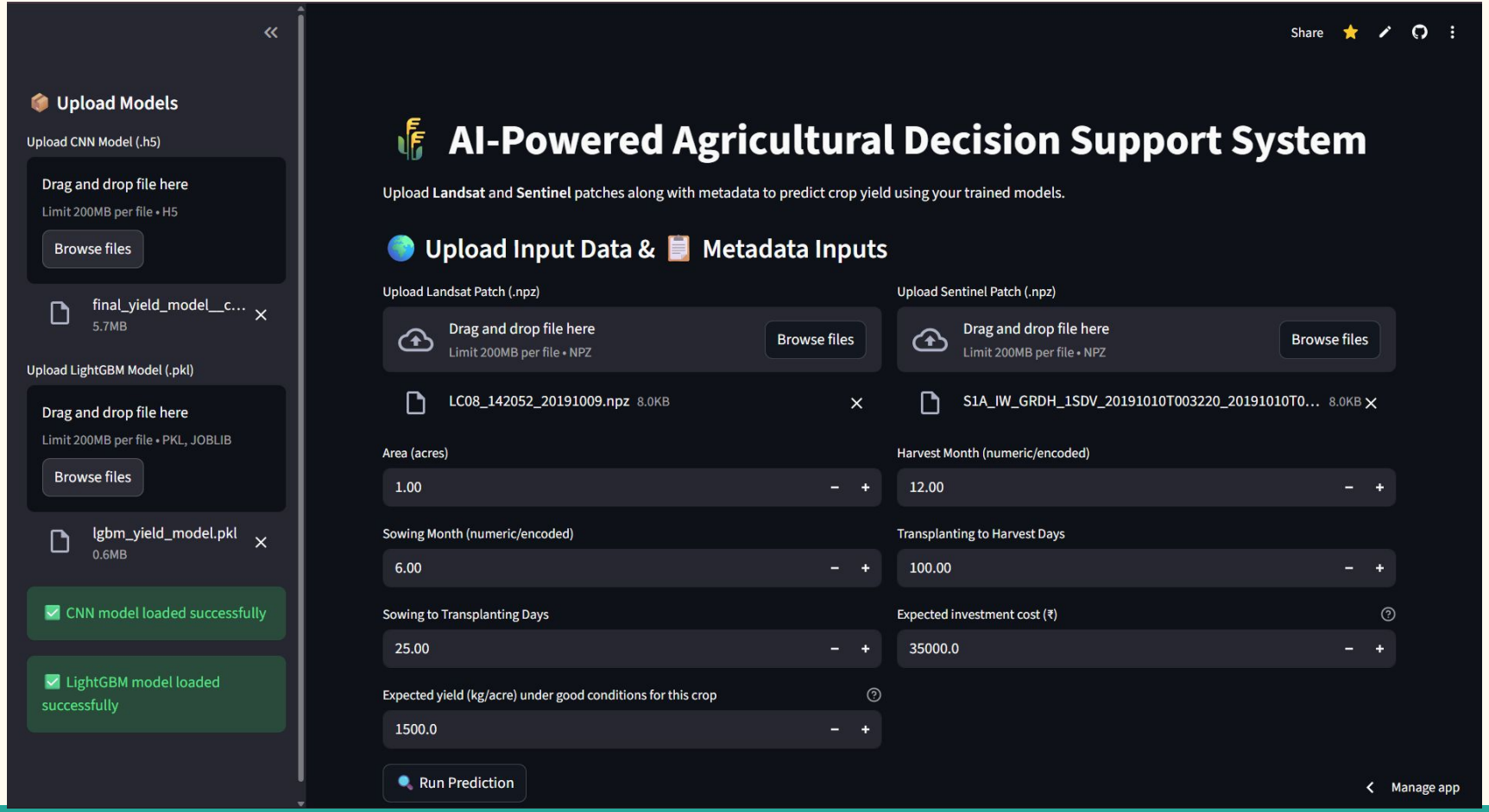


Kaggle link:
Rag + LLM: https://www.kaggle.com/code/sriram150904/agro-llm
Yield Prediction: https://www.kaggle.com/code/sriram150904/fork-of-paddy-model

Demonstration



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- Extracted CNN feature vector shape: `(128,)`
- LightGBM model expects features: `138`
- App is sending features: `138`
- CNN feature vector length: `128`

Predicted Yield: 1616.53 kg/acre

Feature Summary

AREA	sow_mon	har_mon	sow_to_trans_days (log1p)	trans_to_har_days	VV_mean	VH_mean	VH_VV_ratio	VH_VV_ratio_trans2	NDVI	CNN_features_dim
1	6	12	3.2581	100	-7.623	-14.742	2.1597	2.1597	0.0672	128

Yield & Economic Interpretation | LLM Advisory (HF Router) | Live AI Q&A | Agro RAG Knowledge Bot

Yield and Economic Analysis

Predicted yield: 1616.53 kg/acre (107.8% of expected)
Total area: 1.00 acres
Predicted total yield: 1,616.5 kg
Market price used: ₹25.00/kg
Predicted total revenue: ₹40,413
Total investment cost: ₹35,000

⚠️ **Low Profit:** ₹5,413 (15.5% margin)

Yield Assessment: Good — near expected performance. (Predicted: 1616.53 kg/acre)

Manage app

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📈 Yield & Economic Interpretation 🌿 LLM Advisory (HF Router) 💬 Live AI Q&A Agro RAG Knowledge Bot

🌿 AI-Powered Agronomic Advisory

✅ AI Advisory Generated

My Field Visit to Your Paddy Plot

Walking through your 1-acre field in the Kaveri delta, I can see this is a **short-duration (25 days) nursery-raised crop**—likely sown in June and transplanted before the monsoon peak. The plants are now nearing the **early grain-filling stage** (around 90-100 days post-transplant), which is a critical time for yield determination. Right now, the crop is underperforming.

The **low NDVI value of 0.067** suggests the canopy is **very weak**, almost bare. This could mean the plants are **stunted, stressed, or poorly established**, possibly due to **nutrient deficiency, moisture stress during transplanting, or a lack of healthy tillering**. The **VV and VH radar readings** (-7.623 and 2.160 respectively) further indicate **low biomass and poor canopy structure**—the crop isn't growing as densely or vigorously as it should at this stage.

Given the **Kaveri delta's typical conditions**, this could be because:

- The nursery was poorly managed, leading to weak seedlings.
- The transplanting was done in **low moisture or during a dry spell**, causing shock.
- The field might have had **excessive waterlogging or flooding** right after transplanting, suffocating roots.
- **Nitrogen stress** is common at this stage if the crop didn't establish well quickly.
- A **pest or disease outbreak** (like stem borer or sheath blight) may have weakened the plants early on.

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📈 Yield & Economic Interpretation 🌿 LLM Advisory (HF Router) 💬 Live AI Q&A Agro RAG Knowledge Bot

💬 Ask the Agronomic AI (Live)

Ask your question about this land (Paddy – Kaveri Delta specific):

can i grow paddy crops in this plot?

Generate Answer

✅ AI Response

No, you **should not** grow paddy in this plot right now—it's not suitable for a **Kharif season (June sowing) paddy crop** in the Kaveri delta.

Your **NDVI (0.067)** is very low, meaning the soil is dry and possibly nutrient-poor or heavily stressed. The **radar ratio (2.160)** suggests dense, compacted soil that might struggle with water retention. Meanwhile, the **predicted yield of just 1,616 kg/acre** (far below the Kaveri delta's typical 2,500–4,000 kg/acre for Kharif paddy) confirms the poor growing conditions.

What's Happening in the Field?

- The land is **severely dry**—NDVI this low usually means the plot hasn't received proper pre-sowing irrigation or rainfall.
- The **VH/VV ratio** suggests the soil may be **clay-heavy and compacted**, which can cause poor aeration, drainage issues, or waterlogging if not managed properly.
- If paddy is planted now without **corrective action**, you risk **low

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Agricultural Knowledge Assistant (RAG Powered)

Ask any agriculture-related question:

how to prevent soil erosion?

Search Knowledge Base

Top Retrieved Answers:

Result 1

Q: How can I prevent soil erosion on my farm?

A: planting cover crops, using conservation tillage practices, and building contour terraces. It's also important to avoid overgrazing and to maintain vegetation along streams and other waterways to prevent bank erosion.

Result 2

Q: How can I prevent soil erosion on my farm?

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Result 3

Q: How can I prevent soil erosion on my farm?

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Phase 1: ML + DL + LLM

🌾 Hybrid Yield Prediction & AI Advisory System

🛰️1️⃣ Vegetation Feature Extraction (CNN)

- Used Landsat satellite image patches
- Computed NDVI and spectral bands
- Trained custom CNN to extract 128-D feature vector
- Captured spatial crop health patterns

📊2️⃣ Hybrid Yield Prediction (LightGBM)

- Combined:
  - CNN feature vector
  - NDVI & radar indicators
  - Tabular farm inputs (area, season, duration)
- Trained LightGBM regressor on fused features
- Predicted log yield → converted to actual yield

💰3️⃣ Economic Analysis

- Calculated total production from predicted yield
- Integrated farmer investment inputs
- Estimated profit/loss and margin percentage

🤖4️⃣ Explainable Agronomic Advisory (LLM)

- Injected yield + NDVI + radar + season data into prompt
- Used Mistral LLM (HF Router)
- Generated field-level interpretation
- Provided risk level, stress analysis, and actionable steps

Phase 2: GenAI - RAG & LLM

🌾 RAG-Based Agricultural QA System (FAISS + Qwen 1.8B)

📚 Dataset Preparation

- Combined multiple agricultural QA datasets (~92K)
- Cleaned text and removed duplicates
- Standardized into question–answer format

🧠 Embedding Generation

- Used **SentenceTransformers (all-MiniLM-L6-v2)**
- Converted all questions into 384-dimensional dense vectors
- Generated embeddings for semantic similarity search

🗂️ Vector Database (FAISS)

- Built FAISS IndexFlatL2 for fast similarity search
- Stored ~92K embeddings
- Enabled Top-K semantic retrieval

🔎 Retrieval Step

- Encoded user query into embedding
- Retrieved top 5 most similar QA entries
- Passed only retrieved knowledge to generator

🤖 Generative Model

- Used **Qwen 1.8B Chat model**
- Constructed strict grounding prompt
- Enforced rule: use only retrieved knowledge
- Generated structured agricultural answer

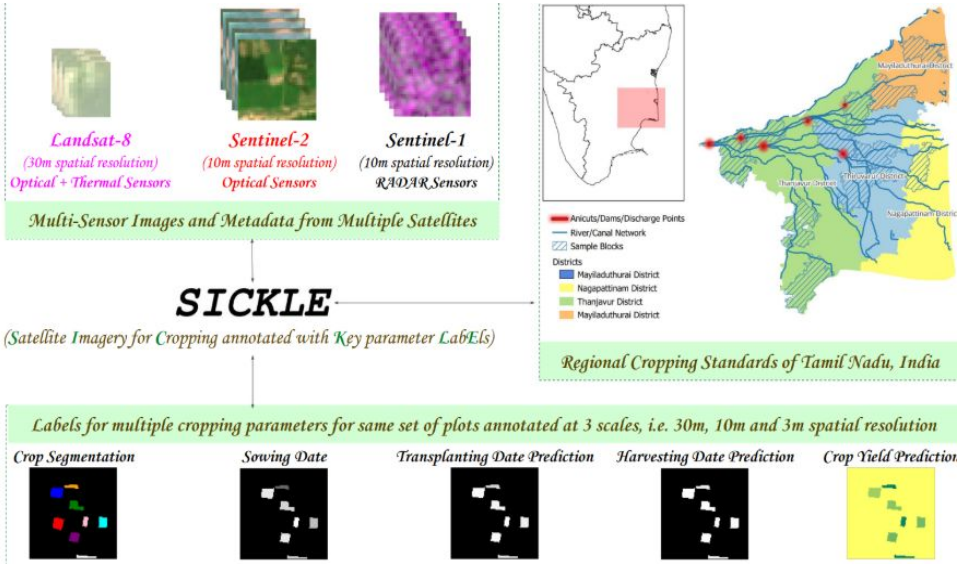
🎯 Outcome

- Reduced hallucination
- Improved domain grounding
- Enabled scalable, offline GenAI system on Kaggle

Results Comparison of Yield Prediction

Existing Previous Work Results

Task	Metric	L8
Crop Type (SI)	IoU (%)	$47.73\% \pm 1.77\%$
Crop Type	IoU (%)	$56.04\% \pm 5.84\%$
Sow Date	MAE (days)	2.66 ± 0.961
Transplant Date	MAE (days)	$\mathbf{6.20 \pm 1.030}$
Harvest Date	MAE (days)	$\mathbf{9.86 \pm 0.736}$
Crop Yield (SI)	MAPE (%)	$\mathbf{46.74\% \pm 3.82\%}\%$
Crop Yield (RS)	MAPE (%)	$\mathbf{54.00\% \pm 9.67\%}$
Crop Yield (AS)	MAPE (%)	$\mathbf{59.38\% \pm 14.75\%}$



My Work results

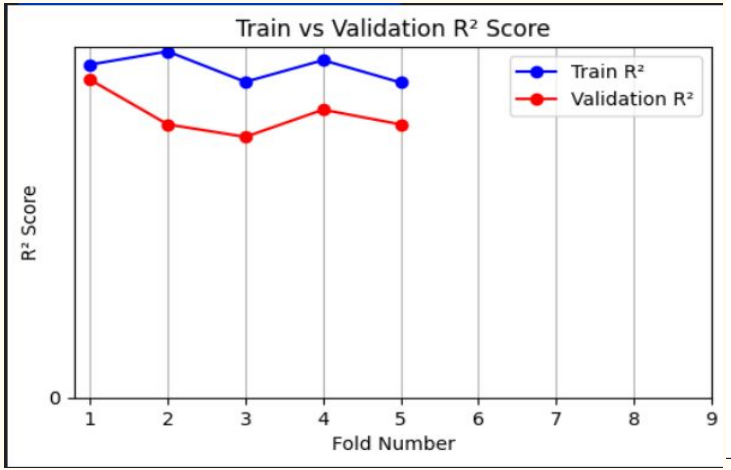
Task: Crop Yield (SI)

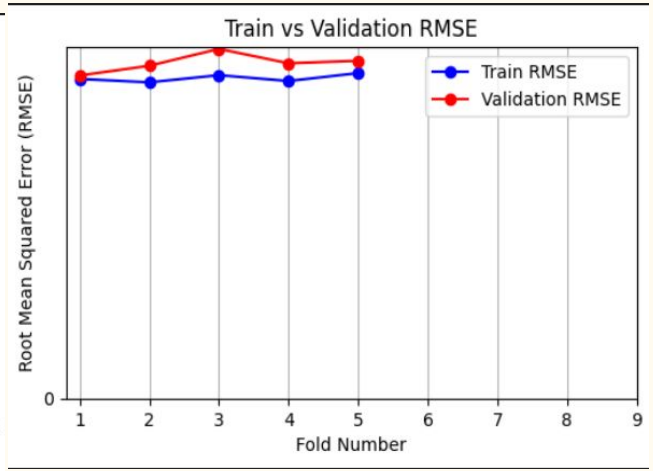
Train set MAPE (%) = 13.01%
Test set MAPE (%) = 15.91%

MAPE - Mean Absolute Percentage Error

Accuracy = 100 - MAPE = 85%

Cross Validation Results





- **RMSE Plot:**
  - Train RMSE is slightly lower than validation RMSE.
  - Both are around **0.17 to 0.18** on the log scale.

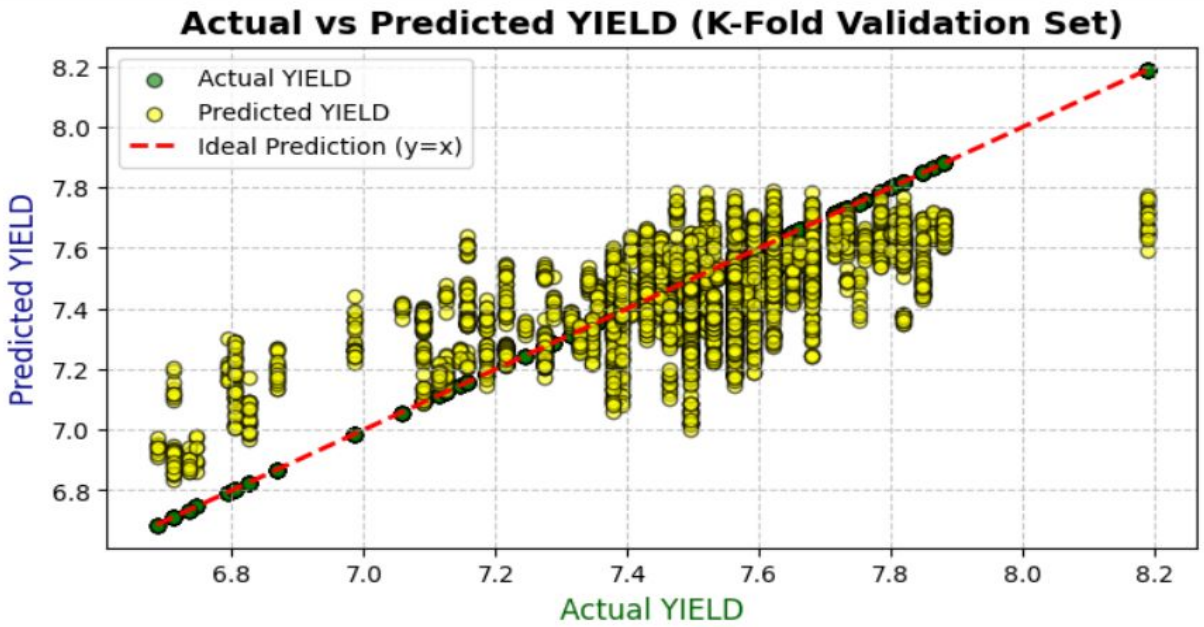
What it shows:

- **R^2 Plot:**
  - Train R^2: ~0.40–0.45 across folds.
  - Validation R^2: ~0.32–0.38 across folds.
  - **Gap of ~0.05–0.07 between train and validation R^2.**

Insights:

- **Consistent results across folds**, no extreme variations.
- Slight mild overfitting is present (**Train R^2 > Validation R^2 by ~0.05–0.07**). **Acceptable** in real-world regression tasks
- The error is relatively stable across folds, indicating the model is robust and **not sensitive to data splits**.

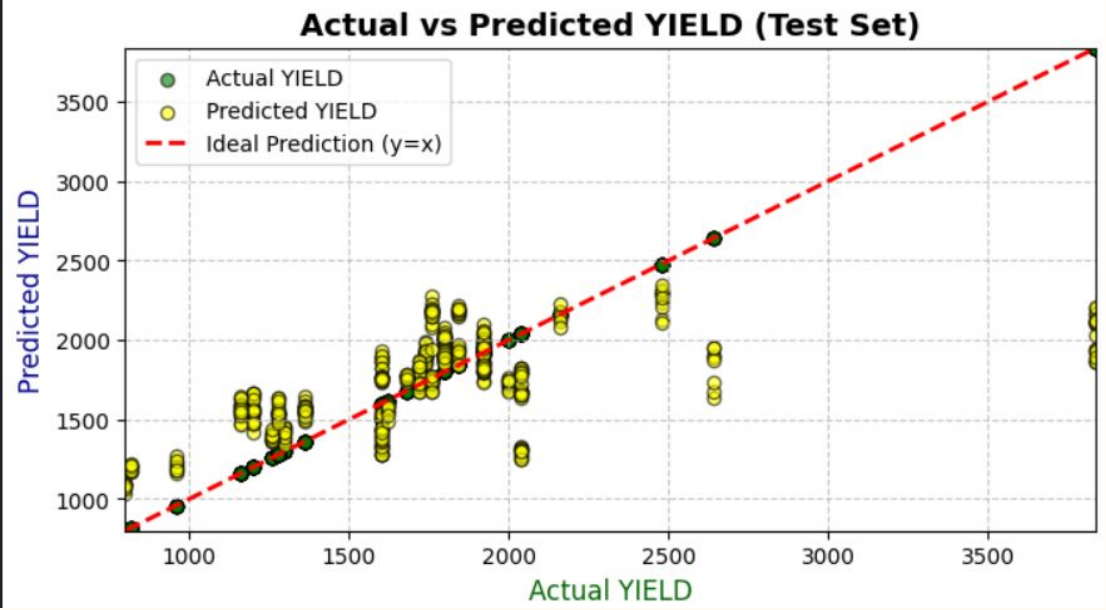
Model Performance Analysis – LightGBM (Paddy Yield Prediction)



- The model's predictions **closely follow the ideal line**, especially in the mid-range yields (~7.0 to 7.8).

Insights:

- Some **underprediction and overprediction are visible**, especially at the lower yield values (~6.8–7.2), showing a slight bias.
- Overall, the scatter spread is **reasonable for real-world yield data**, indicating that the model generalizes decently but not perfectly.



What it shows:

- On the **original scale** after applying `expm1()` to reverse the log transformation.
- The yield values range from **~800 to ~3500 units.**
- Model predictions mostly follow the ideal line, but the error range is visibly larger than in the log scale.

Sample Output

Insights:

- The model predicts **large-scale yield variations moderately well**, but there are significant differences for some test cases.
- This plot highlights the **difficulty of modeling raw yield values**, especially due to **variance** in high-yield samples.

Yield and Economic Analysis

Predicted yield: 1616.53 kg/acre (107.8% of expected)
Total area: 1.00 acres
Predicted total yield: 1,616.5 kg
Market price used: ₹25.00/kg
Predicted total revenue: ₹40,413
Total investment cost: ₹35,000

⚠ **Low Profit:** ₹5,413 (15.5% margin)

Yield Assessment: Good — near expected performance. (Predicted: 1616.53 kg/acre)

GenAI - RAG & LLM

```python
model_embed = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")

corpus = final_df["question"].tolist()

embeddings = model_embed.encode(
    corpus,
    show_progress_bar=True,
    convert_to_numpy=True,
    batch_size=64
)
```

```python
import faiss

dimension = embeddings.shape[1]

index = faiss.IndexFlatL2(dimension)
index.add(embeddings)

print("FAISS index size:", index.ntotal)
```

```
FAISS index size: 92385
```

```python
def retrieve(query, top_k=5):
    query_embedding = model_embed.encode([query])
    query_embedding = np.array(query_embedding).astype("float32")

    distances, indices = index.search(query_embedding, top_k)

    results = final_df.iloc[indices[0]]

    return results
```

```python
tt = retrieve("How to control aphid infestation in rice?")
```

	question	answer
39409	asking about the control measure to control ap...	suggested to spray nuvacron 40ec@2 mililit
4307	asking about to controll of aphids in rice.	suggested him to spray durseban @ 2ml/lite
45007	asking about the control of aphid in rice.	recommended to apply tricel 20 ec @ 2ml p
51954	asking about the control measure to control ap...	suggested to apply furadon3g@3gram/squa
58559	problem of aphids in boro rice	advised him to apply rogor 35 ec @ 2 ml / lit

```python
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

model_name = "Qwen/Qwen1.5-1.8B-Chat"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model_gen = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float16,
    device_map="auto"
)
```

```
print(generate_answer("What is the fertilizer dose for cabbage?"))
```

```
The fertilizer dose for cabbage can be recommended using the following knowledge entries:

1. Urea: The recommended amount of urea for cabbage is 36 kg/bigha in a split dose, which means app
lying it in two separate applications. This dosage is spread evenly across the cabbage plants and h
elps to promote root growth, enhance nutrient uptake, and prevent wilting.

2. Ssp (Supplementary Spraying): To provide additional nitrogen to the cabbage plants, a supplement
sprayer can be used at a rate of 50 kg/bigha. This application should be applied immediately after
the basal dose of urea, ensuring that all cabbage leaves receive adequate nitrogen support.

3. Mop: A mop can also be added to the basal dose of urea to help with water retention and reduce s
urface water evaporation. The mop should be applied at a rate of 13 kg/bigha, ensuring that the ent
ire cabbage bed receives sufficient moisture.

4. Borax: Borax is a mineral that can be used to improve soil fertility by increasing pH levels and
promoting beneficial microorganisms. It can be applied at a rate of 2.5-3 kg/bigha, depending on th
e specific needs of the cabbage plants.
```

💬 Ask the Agronomic AI (Live)

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Generate Answer

✅ AI Response

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🚀 Final Outcome

An integrated system that:

✔ Predicts crop yield
✔ Estimates economic performance
✔ Explains model outputs
✔ Provides grounded, domain-specific advisory

Thank You

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